

SAMUEL RYOO

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EDUCATION

University of California, Irvine

Bachelor of Science, Computer Science and Engineering

Irvine, CA

Expected June 2027

- GPA: 3.608; Dean's Honor List, 6 quarters; Rose Hills Foundation Undergraduate Scholarship
- Relevant Coursework: Embedded Software, Digital Systems, Digital Logic Lab, Organization of Digital Computers, Circuit Analysis, Electronic Devices and Circuits

SKILLS

Programming: C, C++, Embedded C, Python, Verilog

Embedded Systems: Microcontrollers (ATmega32, ESP32, STM32, Arduino), bare-metal register-level firmware, interrupt-driven design, hardware timers, real-time systems, sensor integration

Protocols & Tools: UART, I2C, PWM, GPIO, KiCad, Vivado, Cadence Virtuoso, Git, ArduPilot, EdgeTX, Mission Planner

Test & Debug: Hardware bring-up and fault isolation, serial-link debugging, sensor calibration, self-checking testbenches

EXPERIENCE

Refined Robotics

Hardware Engineering Intern

Tokyo, Japan

June 2026 - September 2026

- Wrote Arduino motor-control firmware generating STEP/DIR pulse sequences for three DRV8825-driven stepper axes in a delivery mechanism that replaced an open-loop servo design, with motion commands received over USB serial.
- Designed a second-revision board instrumenting every output branch with Kelvin shunts and INA3221 monitors, aggregating per-branch voltage and current telemetry on an STM32G031 over I2C and reporting to the host over UART.
- Integrated LiDAR, stereo cameras, GNSS and onboard Jetson compute at the hardware level, wiring Ethernet, USB and GMSL2 links and building the connector and harness interfaces for the sensor and compute payload.
- Designed a two-layer power distribution PCB in KiCad routing 12V to five sensor, compute and communication subsystems, passing ERC and DRC with zero violations and releasing it to fabrication and assembly.

Making an Impact

Founder and President

Los Angeles, CA

September 2020 - August 2025

- Founded and led a certified non-profit for 5 years, organizing recurring homeless outreach and fundraising events that delivered care packages to people in need, and growing it to 200+ members across 10 schools and 4 districts.

PROJECTS

AVR-Based Embedded Memory Game

Embedded Systems Project (Team of 2)

Irvine, CA

August 2025 - October 2025

- Programmed an ATmega32 bare-metal in embedded C, configuring GPIO through DDRx, PORTx and PINx registers with bit masking to drive a matrix keypad, LEDs, a character LCD and a speaker without disturbing unrelated pin state.
- Architected interrupt-driven firmware around a non-blocking main loop so keypad input, LCD rendering and tone generation were serviced concurrently without any peripheral monopolizing the CPU.
- Built deterministic millisecond timing on Timer0 with a prescaler, polling the overflow flag rather than instruction-count delay loops so timing tracked the crystal oscillator regardless of code path.

Multirotor UAV Avionics and Embedded Systems

Embedded Systems / Controls Project (Team of 4)

Irvine, CA

September 2025 - December 2025

- Remapped the flight controller's UART assignment in ArduPilot (BRD_ALT_CONFIG) to route CRSF through SERIAL6, aligning baud rates and protocol settings across radio, autopilot and ground station.
- Configured a MAVLink telemetry bridge on an ESP32 running DroneBridge firmware, streaming packets at 57.6 kbaud over WiFi to a Mission Planner ground station for real-time flight monitoring.

EMG-Controlled Robotic Hand

Embedded Biosignal Acquisition and Actuation

Personal Project

August 2026 - Present

- Building an EMG-controlled robotic hand with a MyoWare sEMG front end sampled over I2C by an ADS1015 ADC and real-time DSP on an ESP32, with bandpass, 50/60 Hz notch, rectification and envelope detection driving servo actuation.
- Designed a two-rail power architecture isolating servo transients from the biosignal front end through a separate 5 V supply and single star ground, with BLE streaming to a Python plot for live signal inspection.

INTERESTS

Ad Hoc Cooking, Golf, Dodgers Baseball, Aconstic Guitar, Backpacking the San Gabriel Mountains